

[I225] Statistical Signal Processing(E) Office Hour 5

1. Suppose the following Bernoulli observations are obtained:

$$\mathbf{x} = [1, 0, 1, 1, 0].$$

Each observation follows

$$X_i \sim \text{Bernoulli}(p), \quad i = 1, 2, \dots, N,$$

where the parameter represents the probability of success.

- (a) Derive the likelihood function.
- (b) Derive the log-likelihood function.
- (c) Find the maximum likelihood estimator.
- (d) Explain why log-likelihood is commonly used.

Answer:

(a)

Since the observations are independent, the likelihood function is

$$L(p|\mathbf{x}) = \prod_{i=1}^N p^{x_i} (1-p)^{1-x_i}.$$

For the given observations, there are three successes and two failures. Therefore,

$$L(p|\mathbf{x}) = p^3 (1-p)^2.$$

(b)

Taking the logarithm gives

$$\ell(p) = \log L(p|\mathbf{x}) = 3 \log p + 2 \log(1-p).$$

(c)

Differentiate the log-likelihood with respect to the parameter:

$$\frac{d\ell(p)}{dp} = \frac{3}{p} - \frac{2}{1-p}.$$

Set the derivative equal to zero:

$$\frac{3}{p} = \frac{2}{1-p}.$$

Solving this equation gives

$$3(1 - p) = 2p, \quad 3 = 5p.$$

Hence,

$$\hat{p} = \frac{3}{5} = 0.6.$$

(d)

Log-likelihood is commonly used because it converts products into sums. This makes differentiation and numerical optimization easier, while the maximum point remains unchanged.

2. Suppose the observed random vector is

$$\mathbf{X} = [X_0, X_1, \dots, X_{N-1}]^T,$$

where each sample follows

$$X_n \sim \mathcal{N}(\mu, \sigma^2), \quad n = 0, 1, \dots, N-1.$$

Assume that the variance is known.

- (a) Derive the likelihood function.
- (b) Derive the log-likelihood function.
- (c) Find the maximum likelihood estimator of the mean.
- (d) Explain the meaning of the obtained estimator.

Answer:

(a)

Since all samples are independent and identically distributed, the likelihood function is

$$p(\mathbf{x}; \mu) = \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x_n - \mu)^2}{2\sigma^2} \right].$$

(b)

Taking the logarithm gives

$$\ell(\mu) = \log p(\mathbf{x}; \mu) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x_n - \mu)^2.$$

(c)

Differentiate the log-likelihood with respect to the mean:

$$\frac{\partial \ell(\mu)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x_n - \mu).$$

Set the derivative equal to zero:

$$\sum_{n=0}^{N-1} (x_n - \mu) = 0.$$

Therefore,

$$\hat{\mu} = \frac{1}{N} \sum_{n=0}^{N-1} x_n.$$

(d)

The maximum likelihood estimator of the mean is the sample mean. This means that the value making the observed data most likely is the average of all observations.

3. Suppose the random variable follows a Poisson distribution:

$$X \sim \text{Poisson}(\lambda),$$

with probability mass function

$$p(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Assume that independent observations are collected.

- (a) Verify the regularity condition.
- (b) Compute the Fisher information.
- (c) Find the Cramer-Rao lower bound.
- (d) Explain the relationship between Fisher information and estimation accuracy.

Answer:

(a)

For one observation, the log-likelihood is

$$\ell(\lambda) = x \log \lambda - \lambda - \log(x!).$$

The score function is

$$\frac{\partial \ell(\lambda)}{\partial \lambda} = \frac{x}{\lambda} - 1.$$

Taking expectation gives

$$E \left[\frac{\partial \ell(\lambda)}{\partial \lambda} \right] = \frac{E[X]}{\lambda} - 1 = \frac{\lambda}{\lambda} - 1 = 0.$$

Therefore, the regularity condition is satisfied.

(b)

The Fisher information for one observation is

$$I(\lambda) = E \left[\left(\frac{\partial \ell(\lambda)}{\partial \lambda} \right)^2 \right] = E \left[\left(\frac{X}{\lambda} - 1 \right)^2 \right].$$

Since the variance of a Poisson random variable is equal to its parameter,

$$\text{Var}(X) = \lambda,$$

we obtain

$$I(\lambda) = \frac{1}{\lambda}.$$

For independent observations,

$$I_N(\lambda) = \frac{N}{\lambda}.$$

(c)

The Cramer-Rao lower bound is

$$\text{Var}(\hat{\lambda}) \geq \frac{1}{I_N(\lambda)}.$$

Therefore,

$$\text{Var}(\hat{\lambda}) \geq \frac{\lambda}{N}.$$

(d)

Fisher information measures how much information the observations contain about the unknown parameter. Larger Fisher information means a smaller lower bound on variance, so the parameter can be estimated more accurately.

4. An engineer is tracking a drone that is accelerating away from a base station. The position $x[n]$ at discrete time step n is modeled by a quadratic polynomial with three unknown parameters: initial position (θ_1), initial velocity (θ_2), and constant acceleration (θ_3):

$$\hat{s}[n] = \theta_1 + \theta_2 n + \theta_3 n^2$$

We collected 4 noisy measurements at times $n = 0, 1, 2, 3$: $\mathbf{s} = [2, 6, 13, 26]^T$.

Compute the Least Square Estimation (LSE) of $\hat{\boldsymbol{\theta}} = [\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3]^T$ and the final scalar value of the minimum cost function $J(\hat{\boldsymbol{\theta}})$!

Answer:

(a)

Formulate matrix equation

$$\mathbf{s} = \begin{bmatrix} 2 \\ 6 \\ 13 \\ 26 \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

(b)

Compute the LSE of $\hat{\boldsymbol{\theta}}$

$$(\mathbf{H}^T \mathbf{H}) \hat{\boldsymbol{\theta}} = \mathbf{H}^T \mathbf{s}$$

$$\begin{bmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix} \hat{\boldsymbol{\theta}} = \begin{bmatrix} 47 \\ 110 \\ 292 \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} 2.15 \\ 1.15 \\ 2.25 \end{bmatrix} \quad \boxed{\text{by using row elimination or matrix inversion}}$$

(c)

Compute the $J(\hat{\boldsymbol{\theta}})$

$$J(\hat{\boldsymbol{\theta}}) = \mathbf{s}^T (\mathbf{s} - \mathbf{H} \hat{\boldsymbol{\theta}})$$

$$= \begin{bmatrix} 2 \\ 6 \\ 13 \\ 26 \end{bmatrix}^T \begin{bmatrix} -0.15 \\ -0.45 \\ 0.15 \\ 0.95 \end{bmatrix}$$

$$= 23.65$$

5. An engineer models the trend of a signal over a short duration. Initially, they assume the signal can be modeled as a simple linear ramp passing through the origin (Order 1 model):

$$\hat{s}[n] = \theta_1 n$$

Three sequential measurements are collected at times $n = \{0,1,2\}$: $s[n] = \{1,3,5\}$

After applying LSE, the coefficient is estimated as $\theta_1 = 2.6$

Later, to improve model accuracy, the engineer decides to introduce a constant intercept (DC offset) term θ_2 to create an updated, more expressive model (Order 2 model):

$$\hat{s}[n] = \theta_1 n + \theta_2$$

Instead of performing a full batch matrix inversion from scratch for the Order 2 model, use **Order-Recursive Least Squares** to gracefully update the parameters.

Answer:

(a)

Formulate Matrix Equation for Order 1 model

$$\mathbf{s} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \quad \mathbf{H}(1) = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad \boldsymbol{\theta}(1) = [\theta_1]$$

(b)

Order-Recursive Least Square equation

$$1. \text{ Determine: } \mathbf{h}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$2. \text{ Calculate } \mathbf{P}_1^\perp$$

$$\begin{aligned} \mathbf{P}_1^\perp &= \mathbf{I} - \mathbf{H}(1)\{\mathbf{H}^T(1)\mathbf{H}(1)\}^{-1}\mathbf{H}^T(1) \\ &= \mathbf{I} - \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \{5\}^{-1} [0 \quad 1 \quad 2] \\ &= \frac{1}{5} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & -2 \\ 0 & -2 & 1 \end{bmatrix} \end{aligned}$$

$$3. \text{ Calculate } \hat{\theta}(2)$$

$$\begin{aligned}
\hat{\theta}(2) &= \begin{bmatrix} \hat{\theta}(1) - \frac{\{H^T(1)H(1)\}^{-1}H^T(1)\{h_2h_2^T\}\{P_1^\perp s\}}{\frac{h_2^TP_1^\perp h_2}{h_2^TP_1^\perp s}} \end{bmatrix} \\
&= \begin{bmatrix} 2.6 - \frac{\{5\}^{-1}[0 \quad 1 \quad 2] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}}{6} \\ \frac{[1 \quad 1 \quad 1] \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}}{6} \end{bmatrix} \\
&= \begin{bmatrix} 2.6 - \frac{\{5\}^{-1}18}{6} \\ \frac{6}{6} \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 1 \end{bmatrix}
\end{aligned}$$

After increasing the order, the model is updated into:

$$\hat{s}[n] = 2n + 1$$